

RUZAEVKA, Kazakhstan

WMO: 289660

Lat: 52.8224N

Lon: 66.9581E

Elev: 227

StdP: 98.63

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.1	-29.8	-36.4	0.1	-32.9	-33.2	0.2	-29.7	14.4	-8.0	14.2	-8.0	0.9	230	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.0	18.7	30.1	18.3	28.2	17.7	20.3	28.3	19.4	27.3	18.6	26.1	4.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.1	22.8	16.7	12.2	22.1	15.7	11.5	21.6	59.4	28.5	56.3	27.2	53.6	26.3	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	11.5	10.2	-35.4	34.9	2.9	2.0	-37.5	36.3	-39.1	37.5	-40.7	38.6	-42.8	40.0		
(5)			-35.5	22.0	2.8	1.3	-37.6	22.9	-39.2	23.7	-40.8	24.4	-42.9	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-16.2	-15.1	-6.9	5.1	13.4	18.3	19.2	18.0	12.2	4.2	-5.6	-13.4	(6)
(7)		DBStd	14.31	7.27	6.88	6.29	6.20	4.97	4.26	3.56	4.31	4.89	5.04	7.41	7.35	(7)
(8)		HDD10.0	3649	813	703	524	167	24	2	0	1	31	190	467	726	(8)
(9)		HDD18.3	5832	1072	937	783	396	166	52	33	61	193	439	717	984	(9)
(10)		CDD10.0	1036	0	0	0	22	128	249	284	248	96	9	0	0	(10)
(11)		CDD18.3	178	0	0	0	0	11	50	59	49	8	0	0	0	(11)
(12)		CDH23.3	2259	0	0	0	8	213	665	670	576	126	1	0	0	(12)
(13)		CDH26.7	756	0	0	0	1	51	238	218	208	41	0	0	0	(13)
(14)	Wind	WSAvg	3.8	3.8	4.2	4.5	4.0	4.1	3.7	3.2	3.1	3.3	3.6	3.7	4.2	(14)
(15)	Precipitation	PrecAvg	364	18	16	19	26	40	38	60	41	22	30	30	24	(15)
(16)		PrecMax	496	38	31	39	66	98	106	123	114	45	57	53	54	(16)
(17)		PrecMin	222	1	0	2	0	4	6	16	2	3	5	7	4	(17)
(18)		PrecStd	77	9	8	10	17	26	25	32	32	10	13	12	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.4	0.6	7.8	24.3	30.8	33.5	34.1	33.8	31.2	21.7	9.4	1.2	(19)
(20)			MCWB	-0.6	-0.3	4.0	13.9	16.9	19.3	19.2	19.5	17.1	11.9	6.4	0.0	(20)
(21)		2%	DB	-1.7	-1.0	4.0	21.2	27.9	31.4	31.4	31.3	26.9	18.0	6.6	-0.1	(21)
(22)			MCWB	-2.5	-1.9	2.2	12.2	16.3	18.8	19.1	18.6	15.6	10.4	4.6	-0.8	(22)
(23)		5%	DB	-3.3	-2.7	2.4	18.3	25.7	29.5	29.0	29.1	23.5	14.9	4.4	-1.9	(23)
(24)			MCWB	-4.1	-3.4	1.1	10.9	15.2	18.2	18.6	18.0	14.2	8.7	2.9	-2.7	(24)
(25)		10%	DB	-5.2	-4.7	1.1	15.3	23.4	27.2	27.1	26.7	20.6	12.1	2.5	-3.6	(25)
(26)			MCWB	-5.9	-5.5	0.0	9.3	14.1	17.1	18.0	17.3	12.9	7.4	1.2	-4.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.1	4.7	14.8	18.7	21.3	21.6	21.2	17.9	12.7	7.1	0.4	(27)
(28)			MCDWB	0.2	0.5	6.8	23.2	27.8	29.9	28.7	30.3	28.0	19.6	8.6	0.9	(28)
(29)		2%	WB	-2.4	-1.9	2.4	12.7	17.1	19.9	20.5	19.7	16.1	10.9	4.8	-0.9	(29)
(30)			MCDWB	-1.7	-1.2	3.8	19.5	26.1	28.9	27.6	28.5	25.1	16.9	6.3	-0.1	(30)
(31)		5%	WB	-4.1	-3.5	1.1	11.3	15.8	18.8	19.6	18.8	14.6	9.2	3.0	-2.6	(31)
(32)			MCDWB	-3.4	-2.7	2.3	17.6	23.9	27.5	26.6	26.6	22.3	14.1	4.3	-1.9	(32)
(33)		10%	WB	-5.9	-5.5	0.0	9.6	14.6	17.8	18.8	17.8	13.3	7.7	1.3	-4.4	(33)
(34)			MCDWB	-5.2	-4.7	1.1	14.9	22.0	25.7	25.6	25.0	19.9	11.8	2.3	-3.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.8	8.3	8.2	10.6	12.9	12.7	11.5	12.1	11.8	9.3	6.9	7.7	(35)
(36)			MCDBR	7.7	7.4	6.8	16.0	16.6	16.0	15.0	16.2	16.3	15.1	7.7	7.4	(36)
(37)			MCWBR	7.6	7.2	5.8	8.8	7.9	6.8	6.2	6.6	7.8	9.1	6.3	7.2	(37)
(38)		5% WB	MCDBR	7.7	7.2	6.4	14.3	14.8	14.1	12.1	13.9	14.8	13.3	7.3	7.6	(38)
(39)			MCWBR	7.7	7.1	5.5	8.5	7.9	6.5	5.8	6.5	7.5	8.5	6.3	7.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.322	0.376	0.412	0.409	0.407	0.423	0.424	0.370	0.351	0.324	0.290	(40)	
(41)		taud	2.060	2.012	1.955	2.111	2.212	2.294	2.241	2.216	2.315	2.261	2.133	2.045	(41)	
(42)		Ebn at Noon	657	751	784	814	841	846	822	797	800	725	609	571	(42)	
(43)		Edh at Noon	76	109	143	140	133	124	129	125	100	85	71	64	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.23	2.35	3.81	4.70	5.80	6.16	5.73	4.87	3.54	1.89	0.90	0.81	(44)	
(45)		RadStd	0.09	0.13	0.39	0.50	0.47	0.49	0.44	0.44	0.26	0.26	0.11	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page